



## ***Semester I Examinations 2018/2019***

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| <b>Exam Code(s)</b>         | 1SD1/1SD3/1MF1                                                                                                                                                                                                                                                                                                                                                           |
| <b>Exam(s)</b>              | Higher Diploma in Applied Science (Software Design & Development), Higher Diploma in Applied Science (Software Design & Development) Industry Stream, M.Sc. in Software Design & Development                                                                                                                                                                             |
| <b>Module Code(s)</b>       | CT874                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Module(s)</b>            | Programming I                                                                                                                                                                                                                                                                                                                                                            |
| <b>Paper No.</b>            | I                                                                                                                                                                                                                                                                                                                                                                        |
| <b>External Examiner(s)</b> | Prof. Pier Luca Lanzi                                                                                                                                                                                                                                                                                                                                                    |
| <b>Internal Examiner(s)</b> | Prof. Michael Madden<br>* Dr. Séamus Hill                                                                                                                                                                                                                                                                                                                                |
| <b><u>Instructions:</u></b> | <p>Candidates are required to answer:</p> <p><b>Section A (20 Marks)</b> Answer all 10 parts of section A, This section consists of 10 multiple-choice questions all of which should be attempted.<br/><b>Answers are to be written on the MCQ sheet provided, NOT on the Examination paper.</b></p> <p><b>Section B (80 Marks)</b> Answer Any <b>Two</b> Questions.</p> |
| <b>Duration</b>             | 2 hours                                                                                                                                                                                                                                                                                                                                                                  |
| <b>No. of Pages</b>         | 6                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Discipline(s)</b>        | Information Technology                                                                                                                                                                                                                                                                                                                                                   |
| <b>Requirements</b>         | MCQ                                                                                                                                                                                                                                                                                                                                                                      |

**Section A (20 Marks)**  
**Attempt all 10 questions**  
**Mark your answers on the MCQ sheet provided**

- 1) Which of the following methods is not contained within the List class?
- a) hasNext()
  - b) add()
  - c) list()
  - d) remove()
- 2) Which of the following is the correct syntax for creating an array.
- a) `Int[] myArray = new int[10];`
  - b) `[]int myArray = new int[10];`
  - c) `int[] myarray = new [10]int;`
  - d) `int[] myArray = int[10];`
- 3) Polymorphism allows us to \_\_\_\_\_.
- a) have the same message execute a different method depending on the receiving object.
  - b) hide the internal workings of a Java program from the client programmer.
  - c) overload methods and allow the same method name to be used multiple times.
  - d) extend and modify code.
- 4) Assume we have three classes, X Y and Z. Y and Z are siblings and are subclasses of the X class. Which classes have access to a private variable in class X?
- a) X, Y and Z
  - b) X and Y only
  - c) Y and Z only
  - d) X only
- 5) How would we access the 6<sup>th</sup> element of the array, x?
- a) `x(6);`
  - b) `x(5);`
  - c) `x[6];`
  - d) `x[5];`

- 6) The practice of writing multiple constructors to handle different sets of inputs is known as \_\_\_\_\_.
- a) Inheritance
  - b) Overriding
  - c) Overloading
  - d) Multiplicity
- 7) When an array is passed to a method \_\_\_\_\_.
- a) only the reference is passed, and a copy of the array is not created in the method.
  - b) a copy of the array is created within the method.
  - c) the value of the array is passed into the method.
  - d) None of the above.
- 8) What is meant by “returning an object” from a method?
- a) It means you are returning a null value to the caller.
  - b) It means returning a primitive type to the caller.
  - c) It means returning the object’s address to the caller.
  - d) It means returning the object’s value to the caller.
- 9) The use of “this()” \_\_\_\_\_.
- a) is a reference to hello
  - b) is bad style
  - c) is illegal
  - d) is a call to a Function constructor
- 10) Which of the following is not a Boolean operator?
- a) &&
  - b) !
  - c) ||
  - d) ++

**PTO**

**Section B (80 Marks)**  
**Answer Any TWO Questions**

Q 2.

- a) Write an application that:
- i. declares, creates and randomly initialises an *ArrayList* of 10 *Integer* objects
  - ii. sums the contents of the list
  - iii. uses an *enhanced for loop* to display the contents and sum of the list.
- [10]

- b) The application should include a method that writes the contents of the *list* to a text file. The method is to have the following signature:

**public static void** writeToFile(List<Integer> list)

[15]

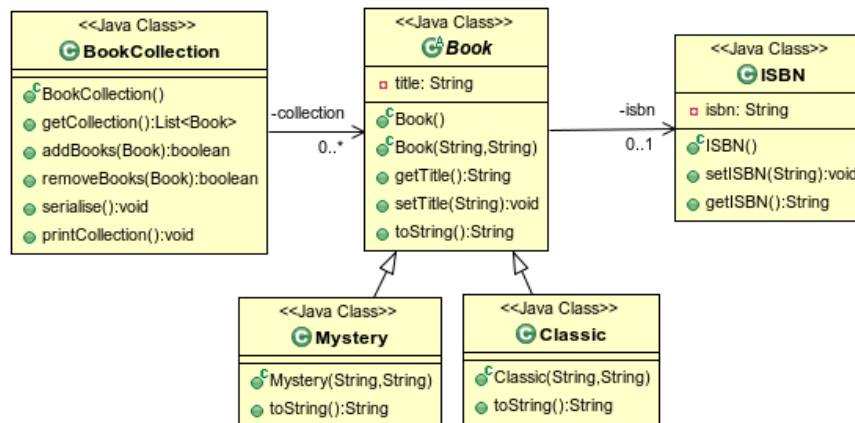
- c) The application should also include a method that reads and displays the contents from the text file. The method signature is as follows:

**public static void** readFromFile(String filename)

[15]

**PTO**

Q 3. From the UML diagram below you are required to:



a) Implement the classes ensuring the classes are *serializable*.

[32 Marks]

b) Write a driver class to carry out the following:

- Create an *LinkedList* and populate the list with an instance of the *Mystery* class and an instance of the *Classic* class.
- Display the contents of the collection.
- Serialize* the List.

[8 Marks]

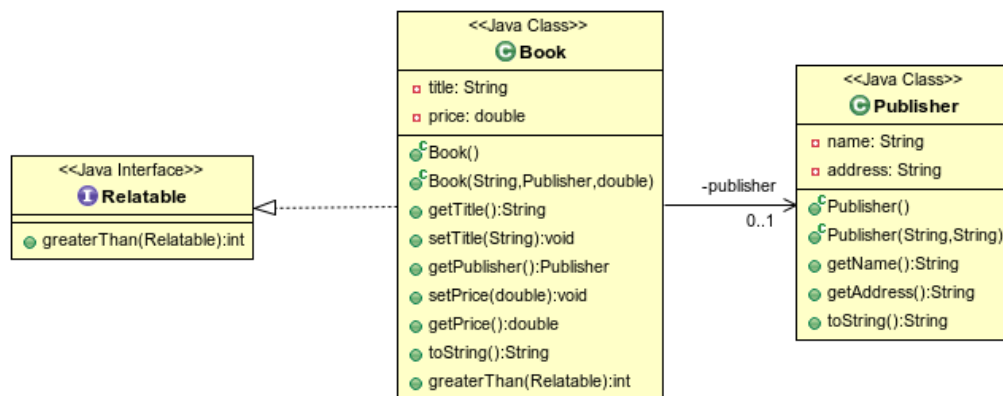
**PTO**

Q 4.

a) In relation to Java describe in your own words, what an *Interface* is.

[5]

b) Using the UML diagram outlined, write a Java application that implements the **Book** class, **Publisher** class and **Relatable** interface to allow comparison between *Book* objects based on *price*.



**Note:** `greaterThan()` returns a 1, 0 or -1 value depending on whether an object is greater than, equal to or less than a similar type of object.

[15]

c) Write a driver class which:

i. Creates and populates an *ArrayList* with three instances of the *Book* class.

[6]

ii. Includes a method with the following signature:

**public static** *Book* findMostExpensive(List<Book> b)

The method uses a *for loop* to traverse the List calling **greaterThan** to determine the most expensive in the List, which is then returned.

[12]

iii. Displays the most expensive book in the List using `toString()`.

[2]